

Experiment No. 2

Basic Chemistry Laboratory Techniques

A) Study of burner

The flame with which wax or oil burns is luminous, i.e. light emitting. Luminous flame introduces soot that acts as a reducing agent, hence the flame is a reducing flame (yellow in color). This flame is rarely used (only for charcoal test) in chemistry laboratory as it does not provide much heat.

In the year 1855, a German scientist Bunsen designed a burner that is in use even today. This burner is known as **Bunsen burner**. It consists of a metal tube with a bore at the base. This bore is surrounded by a metal collar using which, the bore can be opened or closed (Fig. 2.1).

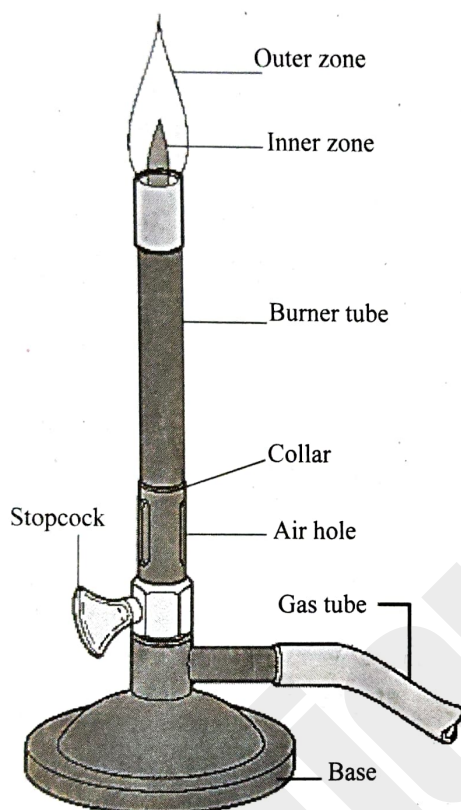


Fig 2.1 Bunsen burner

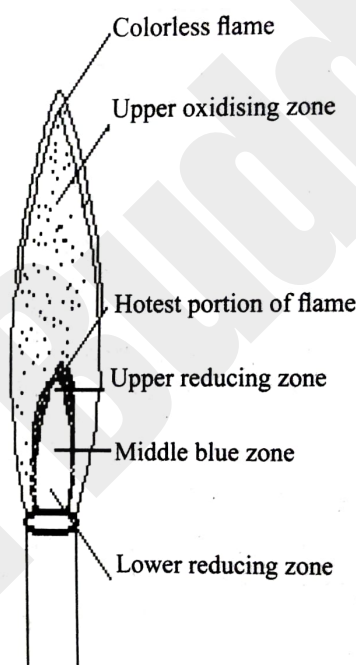


Fig 2.2 Different zones of blue flame

There is a nozzle fitted at the base of the burner. When a jet of fuel gas comes out from the nozzle with the bore open, air rushes into the tube and a mixture of fuel and air burns at the end of the metal tube with a blue flame.

When the gas burns at the end of metal tube with the bore at base closed, combustion takes place in restricted supply of air, resulting in yellow or luminous flame.

The Fig. 2.2 shows different zones of blue flame with which gas burns in Bunsen burner with the bore open. Inner zone of the flame is the hottest zone while lower reducing zone is coldest zone.

B) Operating pinch cock / stopcock

Previously burettes manufactured contained a glass tube calibrated to read volume of solution in mL, which was fitted to a glass nozzle at the bottom with a rubber tube. The rubber tube is closed by a pinch cock. By pressing pinch cock with hand, solution from burette can be added dropwise. However now a days a burette with teflon stopcock is used instead of a rubber tube with pinch cock. (Fig 2.3)

MCQ

Select the correct answer for the following multiple choice type of questions.

1. The flame with which wax or oil burns is called
 - a. luminous flame
 - b. non reducing flame
 - c. oxidizing flame
 - d. blue flame
2. The figure of Bunsen burner flame shows different zones of blue flame, which zone of the flame is the hottest zone?
 - a. Inner zone
 - b. Lower zone
 - c. Outer zone
 - d. Middle zone
3. If acid comes in contact with your skin, then -----
 - a. flush immediately with water
 - b. consult with your instructor
 - c. take help with laboratory assistant
 - d. first remove amount of acid with cloth and then flush with water.
4. Incorrect statement about easy handling of concentrated acids is -----
 - a. storing the acid in glass containers
 - b. using a test tube holder while adding it in a test tube
 - c. using a glass tube with a dropper while adding it in a test tube
 - d. by pipette out with mouth

Short answer questions

1. Write the name of flame used in charcoal cavity test. Why?

The reducing flame is used in the charcoal cavity test because it provides a reducing atmosphere, which helps in the reduction of metallic compounds to metals.

2. Now a days which type of burette is used instead of rubber tube with pinch cock?

Nowadays, a burette fitted with a Teflon stopcock is used instead of a rubber tube with a pinch cock.

3. Name the apparatus when you have to take concentrated acid in a test tube?

A test tube holder is used when handling a test tube containing concentrated acid.

4. How would you dispose concentrated acid in sink after use?

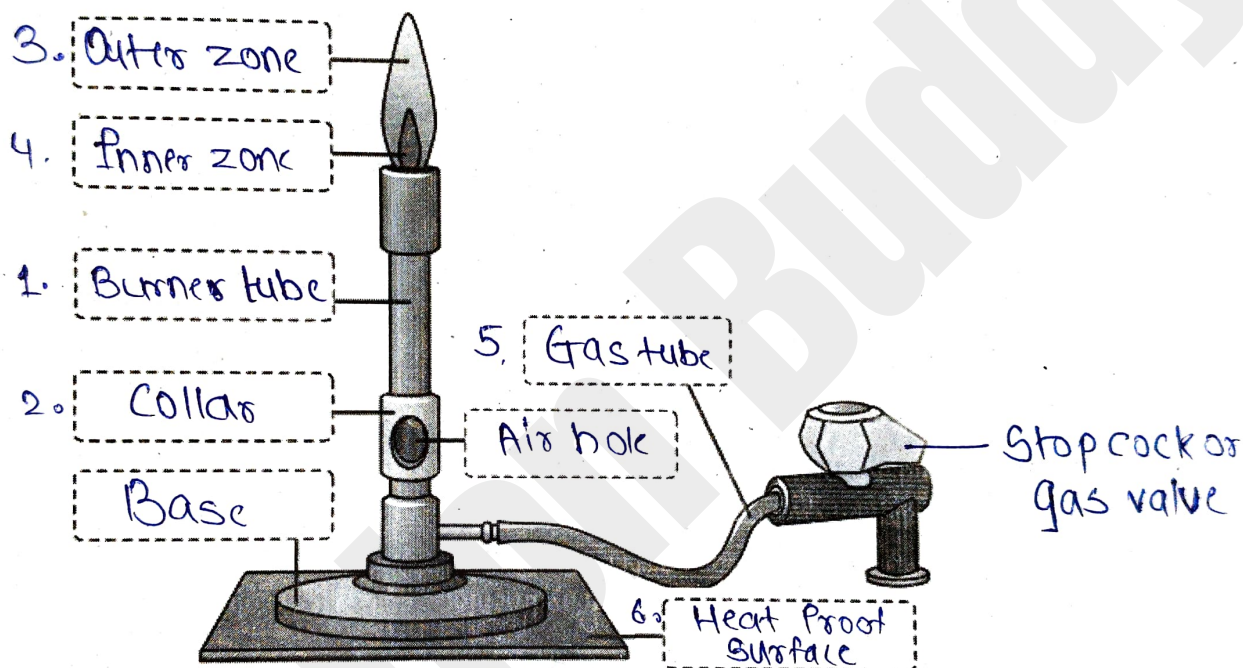
Before disposing of concentrated acid, keep the water tap running and slowly pour the acid into the sink so that it gets diluted and washed away safely with plenty of water.

5. What is the another name to non-sooty (blue) flame?

The non-sooty (blue) flame is also called the oxidising flame.

6. Label the given diagram of Bunsen burner with the phrases given below.

- | | | |
|----------------|-------------|-----------------------|
| 1. Burner tube | 2. Collar | 3. Outer zone |
| 4. Inner zone | 5. Gas tube | 6. Heat proof surface |
| 7. Air hole | 8. Base | |



Remark and sign of teacher: